

[https://github.com/cl-chiu/
NCHC-Agentic-AI-Bootcamp-
2026](https://github.com/cl-chiu/NCHC-Agentic-AI-Bootcamp-2026)

今日教材

- **9:30 ~ 10:50** **Getting Started**

Training

- **9:50 ~ 10:10** NCHC Taiwan AI RAP - Airflow-based
- **10:10 ~ 10:20** NVIDIA NeMo Framework Overview
- **10:20 ~ 10:50** NVIDIA AutoModel

Inference

- **11:00 ~ 11:30** NCHC Taiwan AI RAP - vLLM
- **11:30 ~ 12:00** NVIDIA NIM (Inference Microservice)

Agentic AI Applications

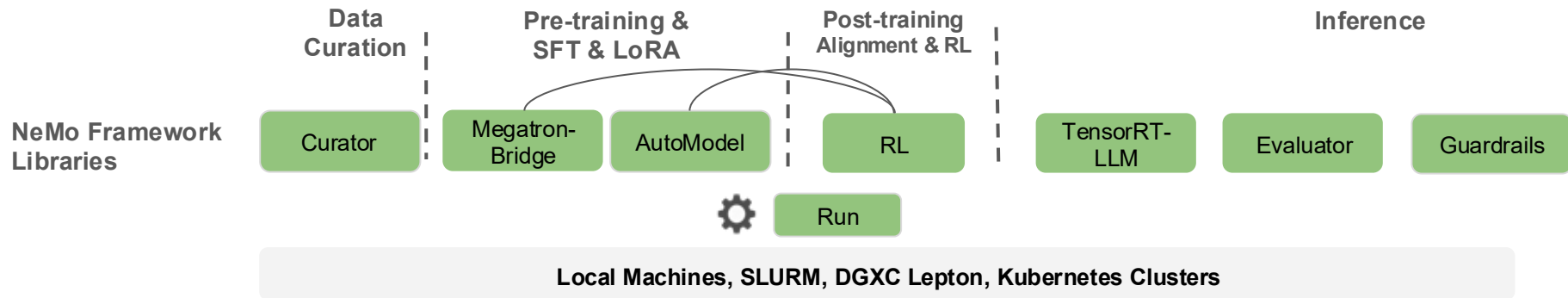
- **13:00 ~ 13:30** NCHC AI RAP - Dify
- **13:30 ~ 14:00** NeMoClaw
- **14:00 ~ 14:10** **Take a break**
- **14:10 ~ 15:30** NVIDIA NeMoAgent Toolkit (NAT) + NVIDIA AI-Q Blueprint
- **15:30 ~ 16:00** Challenge - build your own AI agent



NeMo Framework Overview

Cliff Chiu, Solution Architect | May 2026

Overview of Repos under NeMo Framework



NeMo Curator - Features

Achieve higher accuracy with a variety of GPU-accelerated features



Synthetic Data Generation

- **Pre-built pipelines** - for tasks like prompt generation, dialogue generation, and entity classification
- **Modular** - Easily integrate NeMo Curator's features into your existing pipelines
- **OpenAI API compatible** - Integrate custom Instruct and Reward models



Deduplication & Classification

- **Lexical Deduplication** - Identical (Exact) or near identical (Fuzzy)
- **Semantic Deduplication** - focuses on the meaning rather than the exact text
- **Classifier Models** - State-of-the-art open models to either enrich or filter your data.

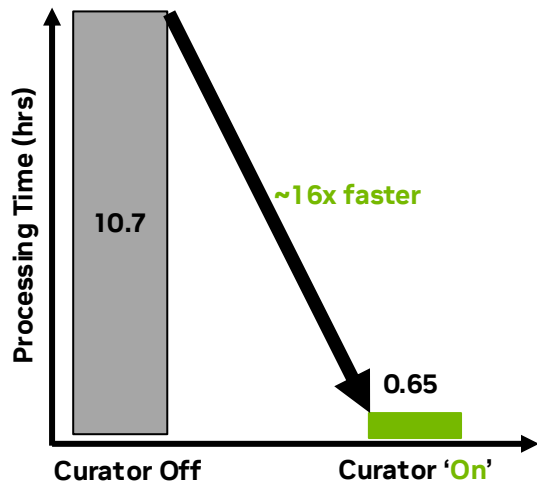


GPU Acceleration with RAPIDS

- **cuDF** - for deduplication & classifier models
- **cuML** - for K-means clustering in semantic deduplication
- **cuGraph** - for fuzzy deduplication

NeMo Curator - Text: World-Class Benchmarks

~16x Faster Processing

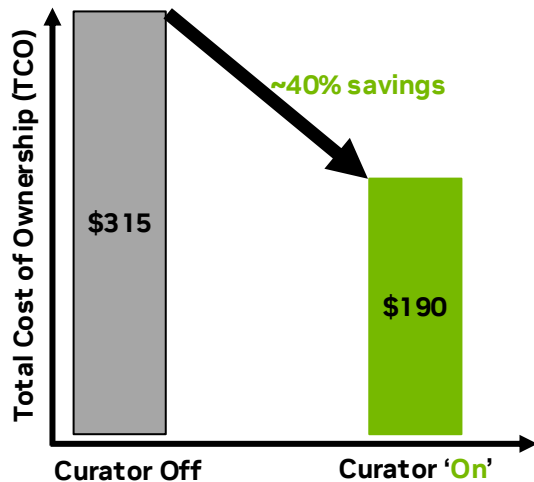


Processing time for fuzzy deduplication of RedPajama-v2 subset (8TB/1.96T tokens)

'On': Data processed with NeMo Curator on 3 H100 nodes

'Off': Data processed with a leading alternative library on CPUs

~40% TCO Savings

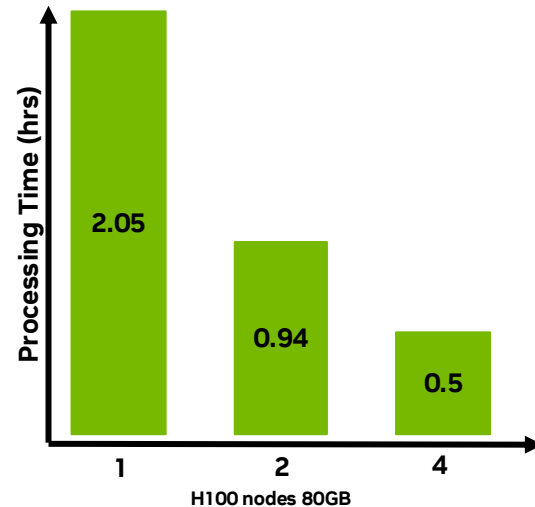


Processing time for fuzzy deduplication of RedPajama-v2 subset (8TB/1.78T tokens)

'On': Data processed with NeMo Curator on 3 H100 nodes

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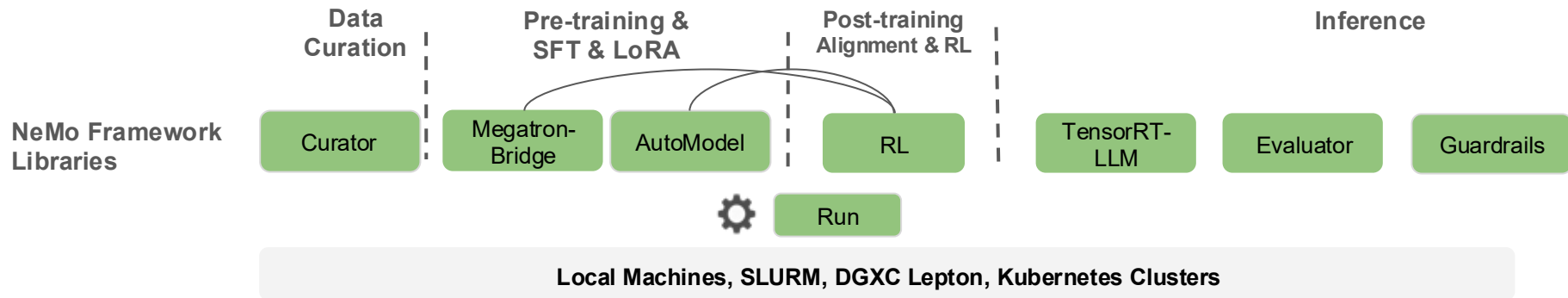
Near Linear Scaling



Processing time for fuzzy deduplication of RedPajama-v2 subset (8TB/1.96T tokens)

Scaling on 1, 2, 4 H100 nodes 80GB

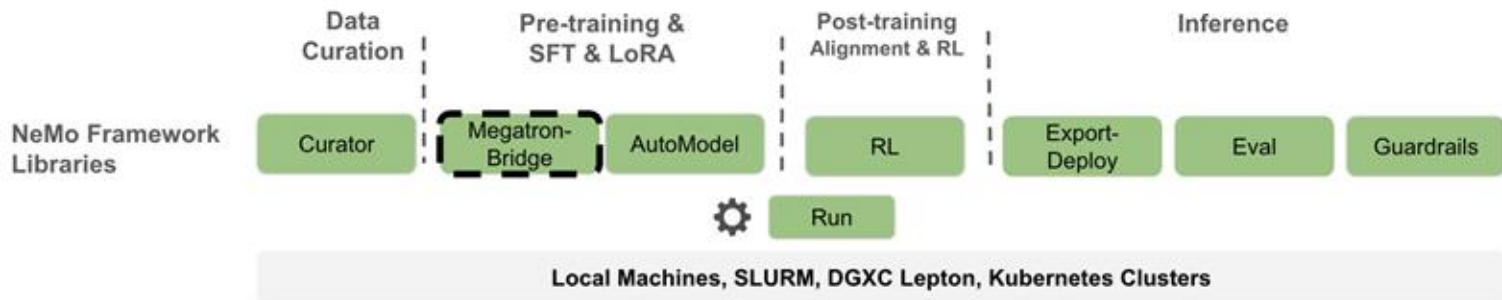
Overview of Repos under NeMo Framework



NeMo Megatron Bridge

What is NeMo Megatron-Bridge?

- Extension library for Megatron-Core for popular community models and optimized recipes
 - Most [optimal benchmark throughput](#) on latest hardware
 - **Modeling** community models ([Supported Models](#))
 - **Bidirectional conversion:** Hugging Face <=> Megatron model
 - Pre-configured high-performance **training recipes** across pre-training / SFT / PEFT
 - **Pre-train, SFT, PEFT (LoRA, DoRA, ...)** support
- PyTorch-native training loop
 - **No PyTorch Lightning** dependency
 - Simplified access for working with Megatron-Core
- Take the best of Megatron and NeMo
 - Performance, Python-first API, easy programmatic setup, support for top models



Current Repo

Supported Models

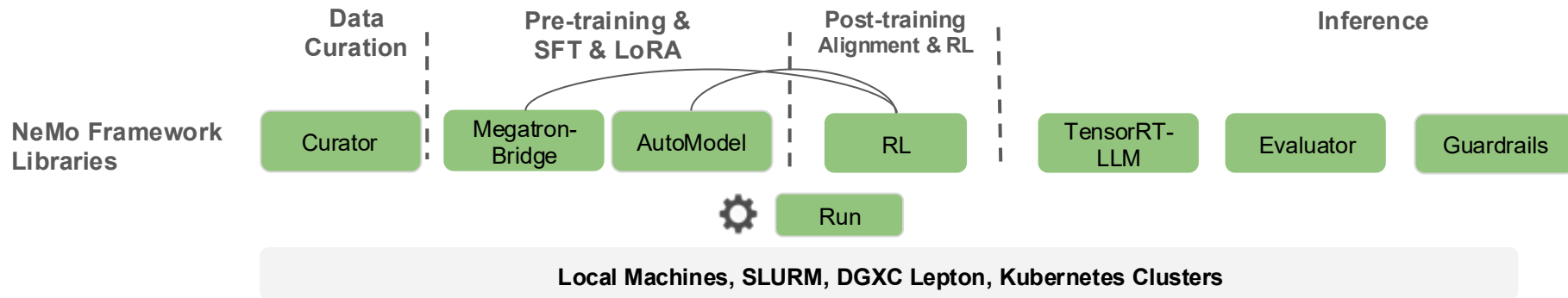
LLMs

- DeepSeek v2/v3
- Gemma
- GLM-4.5
- GPT-oss
- Llama 3, Llama Nemotron
- Mistral
- Nemotron-H, Nemotron Nano v2
- OLMoE
- Qwen 2, Qwen 3, Qwen 3 MoE, Qwen 3 Next
- ...

VLMs

- Gemma 3 VL
- Ministral 3
- Nemotron Nano v2 VL
- Qwen 3 VL, Qwen 2.5 VL
- ...

Overview of Repos under NeMo Framework



NeMo AutoModel

NeMo AutoModel

PyT-based. HF compatible Acceleration. Without Checkpoint Conversion

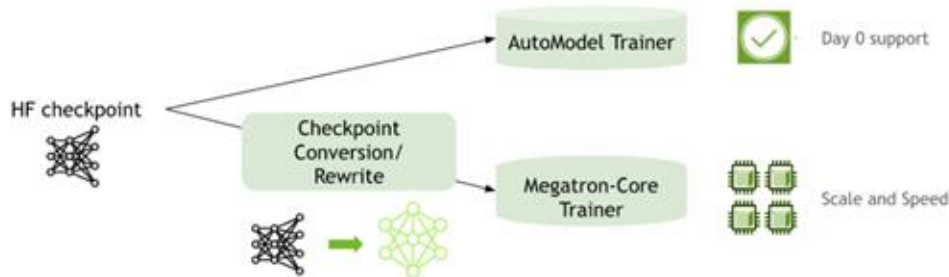
Current Pain points

- **Checkpoint friction:** stability, format conversion, and custom scripts to move from HF to high-performance stacks
- **Model support uncertainty:** “Is model X supported?” and “What changes do I need to make to my code?”
- **Performance gap:** easy HF workflows are convenient but don’t always meet scale/perf requirements as projects grow

How NeMo AutoModel solves that problem:



- **Very easy to use for finetuning LLMs/VLMs, etc**
- **HF-native interface:** Drop-in, HF-style APIs (e.g., AutoModel-like) for model instantiation.
- **Acceleration & Scale:** Uses liger-kernel, FSDP2, DeepEP, full n-d parallelism
- **Fully Interoperable checkpoint** with the rest of the HF ecosystem
- **Transparent:** how to build a good LLM trainer without hiding the details
- **Broad Model coverage** - ([LLM](#), [VLM/OMNI](#))



NeMo Automodel - Balancing between Ease of Use and Speed

Megatron

AutoModel



- **Complex to use.**
- **Limited model coverage**
- **Optimal speed**

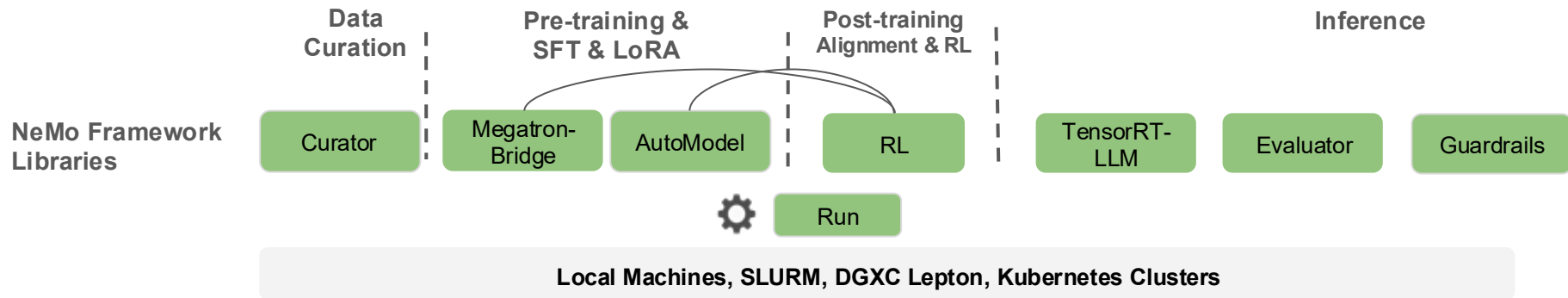
- **Easy to use**
- **Broad model coverage**
- **Decent speed**

- **Easy to use.**
- **Broad model coverage**
- **Lacks Speed**

When to use Automodel vs Megatron-bridge?

	AutoModel	Megatron-Bridge
Parallelisms	n-D parallelism using pyt distributed	n-D parallelism from megatron
Speed	10%~30% slower than Megatron-bridge	Most optimized
Model Coverage	Broad HF model coverage (LLM , VLM/OMNI). Day 0 support for new models	Checkpoint rewrite/conversion needed
Ease of Use & Flexibility	Good and easily debuggable by leveraging PyT	Steeper curve to ramp up

Overview of Repos under NeMo Framework



NeMo RL

Efficient Reinforcement Learning from single to thousands of GPUs

Nemo RL

RL framework for any scale - from single GPU toy experiments to large model post-training.

Algorithms

- SFT
- DPO
- GRPO
 - GRPO LoRA (upcoming)
- DAPO
- On-policy distillation

Infra Components

Training backends

- AutoModel  
- Megatron

Generation backends

- vLLM  
- Megatron inference (upcoming)

Ray for orchestration

Performance

- Any scale: Single GPU to multi-node
- Large MOE (DSV3, Qwen3, GPT-oss) with n-D parallelisms
- FP8 end to end with accuracy matching and 20% speedup. MXFP8 and NVFP4 upcoming
- Blackwell - GB200, B200